



**FEDERAL PUBLIC SERVICE COMMISSION
COMPETITIVE EXAMINATION-2025 FOR RECRUITMENT
TO POSTS IN BS-17 UNDER THE FEDERAL GOVERNMENT**

Roll Number

PHYSICS, PAPER-I

TIME ALLOWED: THREE HOURS	(PART-I MCQs) MAXIMUM MARKS: 20
PART-I (MCQs) : MAXIMUM 30 MINUTES	(PART-II) MAXIMUM MARKS: 80
NOTE: (i) First attempt PART-I (MCQs) on separate OMR Answer Sheet which shall be taken back after 30 minutes.	
(ii) Overwriting/cutting of the options/answers will not be given credit.	
(iii) There is no negative marking. All MCQs must be attempted.	

PART-I (MCQs)(COMPULSORY)

Q.1. (i) Select the best option/answer and fill in the appropriate Box ☐ on the OMR Answer Sheet.(20x1=20)
(ii) Answers given anywhere else, other than OMR Answer Sheet, will not be considered.

1. **The energy of a simple harmonic motion depends upon:**
(A) Mass (B) Frequency (C) Both (A) & (B) (D) None of these
2. **The zeroth law of thermodynamics helps to define the term:**
(A) Temperature (B) Pressure (C) Volume (D) Density
3. **What is the change in the phase if a wave is reflected from a denser medium?**
(A) 3π (B) 0 (C) π (D) 2π
4. **The light can totally pass through the polaroid only if the electric field vector vibrates _____ to the polarizing vector.**
(A) Antiparallel (B) Parallel (C) Opposite (D) None of these
5. **When the net wave intensity is greater than the individual intensities the interference of the waves is said to be.**
(A) Constructive (B) Destructive (C) Both (A) & (B) (D) None of these
6. **The angle of incidence for which polarization is maximum is called:**
(A) Scattering angle (B) Angle of reflection (C) Polarization angle (D) None of these
7. **X-rays are electromagnetic waves of wavelengths of the order of:**
(A) 1nm (B) 0.1 nm (C) 2nm (D) 100 nm
8. **The viscous force the relative motion between the adjacent layers of a fluid in motion. Which one of the flowing fits best in the sentence?**
(A) Opposes (B) Never affects (C) Facilitates (D) May effect under certain conditions
9. **The viscosity of a fluid in motion is 1 Poise. What will be it's viscosity (in Poise) when the fluid is at rest?** (A) 0 (B) 0.5 (C) 1 (D) 2
10. **For the successful operation of heat engine, which condition should be met?**
(A) Cyclic process (B) Operated at certain temperature difference
(C) Both (A) & (B) (D) None of these
11. **Which quantity is not a state function?**
(A) Internal energy (B) Pressure (C) Heat (D) Volume
12. **Fermi-Dirac statistics cannot be applied to:**
(A) Electrons (B) Photons (C) Fermions (D) Proton
13. **In which process all the heat supplied is converted into work done?**
(A) Isothermal (B) Isochoric (C) Isobaric (D) Isentropic
14. **At constant pressure and Volume, exact relation between C_p and C_v is:**
(A) $C_p > C_v$ (B) $C_p / C_v > 1$ (C) $C_p - C_v = R$ (D) All of these
15. **The internal energy of an ideal gas is _____ dependent.**
(A) Pressure (B) Temperature (C) Volume (D) All of these
16. **What is not the condition for the equilibrium in three dimensional system of axis?**
(A) $F_x = 0$ (B) $F_y = 0$ (C) $\sum F_z = 0$ (D) $\sum F \neq 0$
17. **The tendency of rotation of the body along any axis is also called:**
(A) Moment of inertia (B) Moment of couple (C) Torque (D) Force
18. **A simple pendulum is set into oscillations. The bob of the pendulum comes to rest after some time due to:**
(A) Friction of air (B) Its mass (C) Tension in the string (D) Gravity
19. **The energy of a damped oscillator:**
(A) Decreases linearly with time (B) Increases linearly with time
(C) Decreases exponentially with time (D) Increases exponentially with time
20. **The product of mass and velocity is known as:**
(A) Work (B) Moment (C) Impulse (D) Momentum

PART-II

- NOTE:** (i) **Part-II** is to be attempted on the separate **Answer Book**.
 (ii) Attempt **ONLY FOUR** questions from **PART-II**. **ALL** questions carry **EQUAL** marks.
 (iii) All the parts (if any) of each Question must be attempted at one place instead of at different places.
 (iv) Write Q. No. in the Answer Book in accordance with Q. No. in the Q.Paper.
 (v) No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
 (vi) Extra attempt of any question or any part of the question will not be considered.
 (vii) **Use of Calculator is allowed.**

- Q.2.** (a) State and prove Stoke's theorem. (10)
 (b) Prove that if the vector is the gradient of a scalar function then its line integral around a closed curve is zero. (6)
 (c) A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$ where t is the time. Find the components of its velocity and acceleration at time $t=1$ in the direction $2i-3j+2k$ (4) (20)
- Q.3.** (a) Differentiate between Fermi-Dirac, Bose-Einstein and Maxwell-Boltzman's statistics. (10)
 (b) What do you understand by classical statistical mechanics and quantum statistical mechanics? (6)
 (c) A $0.5m^3$ vessel is filled with air at atmospheric pressure. The air is churned by a paddle wheel attached to a shaft 0.1m in diameter, rotating at a speed of 1800 rpm. A force of 5.0N acts on the rim of the shaft. What would be the pressure in the vessel after 10 sec of operation? (4) (20)
- Q.4.** (a) What is viscosity? Discuss effect of temperature on the viscosity of liquids and gases. (10)
 (b) Explain why the level of mercury is down in capillary when placed in container of mercury, while it is up in the capillary in case of water? (6)
 (c) A garden hose has an inside diameter of 2 cm and water flows through it is at 3 m/s. (i) What nozzle diameter is required for the water to emerge at 10 m/s? (ii) At what rate does the water leave the nozzle? (4) (20)
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- Q.5.** (a) Discuss analytical treatment of interference of light to calculate intensity in double slit interference. Describe the conditions of maximum and minimum intensity and also draw the intensity pattern for double slit interference. (10)
 (b) What is resolving power of a grating? Show that it increases with the order of image. (6)
 (c) A body having SHM has an amplitude of 5cm, and a period of 0.2s. When the displacement is 5cm, find the acceleration and velocity. (4) (20)
- Q.6.** (a) What is Carnot Cycle? Draw and explain it for a reversible process. (10)
 (b) What does Gibbs function describe? Prove that Gibbs function for a reversible isothermal and isobaric process is constant. (6)
 (c) A heat engine absorbs 52.4 kJ of heat and exhausts 36.2 kJ of heat in each cycle. Calculate efficiency of heat engine and work done by engine per cycle. (4) (20)
- Q.7.** (a) Derive Clausius – Clayprong equation and also write its two conclusions. (10)
 (b) Prove that $TV^{\gamma-1} = \text{constant}$ (6)
 (c) Air occupying $0.142 m^3$ at 2.04×10^5 is expanded isothermally to 1 atm and then cooled at constant pressure until it reaches its initial volume. Find work done on the gas. (4) (20)
- Q.8.** Write Comprehensive notes on any two of the following: (10 each) (20)
 (a) Newton's ring (b) Polarization by reflection (c) Doppler effect



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PHYSICS, PAPER-II

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PART-I (MCQs)(COMPULSORY)

Q.1. (i) Select the best option/answer and fill in the appropriate Box ☐ on the OMR Answer Sheet.(20x1=20)
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1. **A charge moving in a uniform static magnetic field experiences:**
(A) A force perpendicular to magnetic field and velocity (B) Increase in momentum
(C) Decrease in momentum (D) Decrease in velocity
2. **The same energy alpha particles and electrons associated De Broglie wave lengths are:**
(A) Equal (B) Longer than electrons (C) Smaller electrons (D) None of these
3. **The mass deficit defines:**
(A) Nuclear binding energy (B) Nuclear Quadrapole moment (C) Shape of the nucleus (D) None of these
4. **In nuclear physics the magic number defines:**
(A) Nuclear binding energy (B) Relatively exceptionally stable nuclei (C) Radioactivity (D) None of these
5. **The Zeeman effect is due to the interaction of:**
(A) External magnetic field and total magnetic field of the specimen
(B) External electric field and magnetic field of specimen (C) L, S coupling (D) None of these
6. **A photon having the same energy as an electron has wavelength _____ as electron's De Broglie wavelength.**
(A) Shorter (B) Longer (C) Same (D) None of these
7. **In Laurens transformation motion along _____ axis's remains zero.**
(A) One axis (B) Two axis (C) Three axis (D) None of these
8. **Gausses law is applicable on the charges which are:**
(A) Closely distributed in space (B) Sparsely distributed in space
(C) Enclose by a broken surface (D) Enclose by a close surface
9. **The ratio between the angular momentum and orbital magnetic moment vector of the first orbital electron of Hydrogen Atoms is known as:**
(A) Total Magnetic Moment (B) Bohr Magnetron (C) Magnetic Moment (D) None of these
10. **The cyclotron is a device which primarily used to:**
(A) Accelerate sub atomic particles (B) To generate electrons
(C) To generate neutrons (D) To generate protons
11. **The rate of change of current in a coil is proportional to:**
(A) V (B) - emf (C) R (D) None of these
12. **The charging and discharging behavior of an ideal RLC circuit is:**
(A) Simple harmonic (B) Linear (C) Elliptical (D) Un predictable
13. **The Lorentz force is given by:**
(A) $F_q = q(\mathbf{v} \times \mathbf{B})$ (B) $\Phi = \oint \mathbf{A} \cdot d\mathbf{l} = \oint \mathbf{A} \cdot d\mathbf{l}_x + \oint \mathbf{A} \cdot d\mathbf{l}_y + \oint \mathbf{A} \cdot d\mathbf{l}_z$ (C) $\mathbf{qF} = q(\mathbf{v}_d \times \mathbf{B})$ (D) None of these
14. **The poynting vector represents:**
(A) The rate and direction of electromagnetic energy transferred in space.
(B) The amount of electromagnetic energy transferred in different forms of energy.
(C) Loss of electromagnetic energy in a medium. (D) The flow of thermal energy in a certain direction.
15. **The De Broglie wave particle duality is said to be foundation of uncertainty principle because:**
(A) The velocities of the particles were very high. (B) The mass could not attain the velocity of light.
(C) The mass increases at velocities comparable to the velocity of light increases.
(D) The uncertainty in position of the moving mass.
16. **The relation between half-life and mean life are:**
(A) Inversely proportional (B) Directly proportional (C) Equal (D) Half
17. **The concept of mutual frame of reference was used in:**
(A) Reduced mass correction in spectroscopy (B) Radioactivity
(C) Relative nuclear stability. (D) None of these

PHYSICS, PAPER-II

18. **The Ultraviolet catastrophe means:**
(A) The spectral energy concentration towards longer wave length.
(B) The spectral energy concentration at the middle of the spectrum.
(C) The spectral energy concentration towards shorter wave side. (D) None of these
19. **As per quantum physics, the observables 'E' and _____ can not be simultaneously measured with hundred percent accuracy.**
(A) px (B) t (C) x (D) None of these
20. **The theory of relativity predicts that at velocities comparable to the velocity of light, the moving mass appears to:**
(A) Increase (B) Decrease (C) Remains the same (D) None of these

PART-II

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- Q. 2. (a) Discuss the spectral properties of perfect black body radiation. Explain the ultra violet catastrophe. (10)
(b) Find the fifth & seventh member of Balmer spectral series. If the third member is 1200 Å. (10) (20)
- Q. 3. (a) In quantum mechanics explain the term commutation between the observables. (10)
(b) Explain how De Broglie wave particle duality laid down the fundamentals of uncertainty principle. (10) (20)
- Q. 4. (a) In the context of nuclear physics explain the magic numbers. (10)
(b) Explain nuclear Quadruple moment and explain its significance. (10) (20)
- Q. 5. (a) Derive the expression for electric dipole moment. Discuss two uses in modern science. (10)
(b) Discuss Biot and Savart law and derive its expression, using suitable schematic figure. (10) (20)
- Q. 6. (a) Explain how Hall's effect could be used to find the sign of charge carriers. (10)
(b) Explain the Hall's effect. How it could calculate the number of charge carriers in material. (10) (20)
- Q. 7. (a) Explain the construction and working of photo multiplier tube (PMT). (10)
(b) Compare the energies of a proton and electron having same De Broglie wave length of 1Å. (10) (20)
- Q. 8. Write short notes any TWO of the following: (10 each) (20)
(a) Reduced mass correction in spectroscopy
(b) Electron microscope
(c) Heisenberg uncertainty principal
